ModelIt!

Teacher's Guide

The Water-Energy-Food Nexus: Three Crises, One System

Modeling Coupled Resource Systems Under Population Pressure

Quick Reference

Grade Level	Grade 09L2
NGSS Standard	HS-ESS3-4, HS-ETS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-4, HS-ETS1-2

Evaluate or refine a technological solution that reduces impacts of human activities on natural systems. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students design integrated resource management solutions that account for coupled system dynamics.
Disciplinary Core Idea	**ESS3.A Natural Resources** — Water, energy, and food systems share finite resources, creating trade-offs that require integrated management as demand grows.
Crosscutting Concept	**Systems and System Models** — Students model how coupled subsystems create emergent trade-offs not visible when each system is analyzed in isolation.

CAST Testing Alignment

MULTIPLE CHOICE:

Which intervention would MOST reduce competition between water and energy systems: (a) building more reservoirs, (b) switching to solar/wind energy, (c) importing food, (d) rationing water?

CONSTRUCTED RESPONSE:

Using your model, explain why a city that solves its energy crisis by building natural gas plants might simultaneously worsen its water crisis.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Water, energy, and food are often managed by separate agencies with separate budgets. But they share the same finite resources, and decisions in one system cascade through the others.

Water for Energy

Thermoelectric power plants use 40 percent of US freshwater withdrawals for cooling. Even natural gas plants need water. Only solar PV and wind require minimal water.

Energy for Water

Moving, treating, and heating water consumes 13 percent of US electricity. Desalination is extremely energy-intensive at 3-4 kWh per cubic meter.

Water for Food

Agriculture uses 70 percent of global freshwater. Irrigated agriculture produces 40 percent of food on 20 percent of farmland.

Population Multiplier

Each additional person needs all three resources simultaneously. A city growing 50 percent needs roughly 50 percent more of everything, but the interactions make it more than 50 percent due to coupling.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify water availability, energy production, agricultural output, population demand, industrial usage, and wastewater recycling as nexus components.

E **Establish Relationships**
Students map bidirectional connections: energy needs water, water needs energy, food needs both, population drives all demand.

V **Visualize & Model**
Students build a model showing coupled resource competition and identify leverage points for sustainable management.

E **Evaluate Outcomes**
Students run growth, drought, and recycling scenarios to test how different interventions affect all three systems.

R **Revise & Extend**
Students add desalination, renewable energy, or urban agriculture to explore decoupling strategies.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Your city needs more water, more energy, AND more food. But solving one problem makes another worse. Why?"*

Watch For

Show the nexus concept visually.

Slide 2: Learning Objectives

3 min

Say: *"Today we model why resource crises come in threes."*

Watch For

Pre-teach nexus thinking and resource trade-offs.

Slide 3: Big Question

3 min

Say: *"California built desalination plants for water but needed massive energy. Texas expanded energy but drained aquifers."*

Watch For

Quick-write: How are water, energy, and food connected?

Slide 4: LEVER Framework

2 min

Say: *"Six components, all connected to each other. This is a true system."*

Watch For

Preview the coupled nature of resource competition.

Slide 5: Activity 1: Components

8 min

Say: *"Which of these can city planners directly control?"*

Watch For

Sort external inputs from system responses.

Slide 6: Activity 2: Connect

10 min

Say: *"Every connection goes BOTH ways. Water needs energy, energy needs water."*

Watch For

Map bidirectional relationships. Identify competition points.

Slide 7: Activity 3: Simulate

10 min

Say: *"What happens when the city grows 50 percent? What if drought hits?"*

Watch For

Students test growth, drought, and circular economy scenarios.

Slide 8: Discoveries

5 min

Say: *"You cannot solve water, energy, or food in isolation. They are one system."*

Watch For

Discuss implications for city planning and policy.

Slide 9: STEM Challenge

12 min

Say: *"Your city doubles in 15 years. Design a plan that does not rob Peter to pay Paul."*

Watch For

Groups design integrated resource plans.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Water Availability, Population Demand, Wastewater Recycling
- **INTERNAL:** Energy Production, Agricultural Output, Industrial Usage

Relationships

- **Population Demand to Water Availability:** NEGATIVE (-) — More people consume more water, reducing availability.
- **Water Availability to Energy Production:** POSITIVE (+) — More available water supports power plant cooling and hydroelectric generation.
- **Water Availability to Agricultural Output:** POSITIVE (+) — More water enables more irrigation and crop production.
- **Energy Production to Water Availability:** NEGATIVE (-) — Energy production consumes water for cooling, reducing availability.
- **Agricultural Output to Water Availability:** NEGATIVE (-) — Agriculture is the largest water consumer, depleting supply.
- **Wastewater Recycling to Water Availability:** POSITIVE (+) — Recycling treated water returns it to supply, increasing availability.
- **Population Demand to Energy Production:** POSITIVE (+) — More people drive higher energy demand and production.

Key Discoveries

- Water, energy, and food systems are so interconnected that solving one crisis in isolation can worsen the others
- Energy production is the largest industrial water consumer -- power plants use more water than agriculture in many regions
- Population growth creates compounding pressure because each additional person demands ALL three resources simultaneously
- Circular economy approaches like wastewater recycling can partially decouple the systems and reduce competition

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Water, energy, and food are separate problems"	They share the same finite resources. Energy production is the largest industrial water consumer. Agriculture uses 70 percent of freshwater. Solving one in isolation often makes another worse.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Technology will solve resource scarcity"	Technology can improve efficiency but cannot create resources from nothing. Desalination makes water but needs energy. Solar panels need water to manufacture. Every solution has trade-offs within the nexus.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Population growth creates linear resource pressure"	Population growth creates COMPOUNDING pressure because each person needs ALL three resources, and the coupling means shortages cascade. A 50 percent population increase creates more than 50 percent more demand due to system interactions.